Lecture 11

Saturday, 19 September 2026 11:37 AM

Joint distributions $(X_1, X_2 \ldots X_n)$ — Joint CDF

- Discrete Random Vectors — Joint PMF, Conditional PMF
- Jointly conts R. vectors → Joint PDF, Cond. PDF

Jointly conts Random vectors

X is a conts RV $\iff F_X(x) = \int_{-\infty}^{x} f_X(s)\,ds$ (PDF)

(X, Y) is jointly conts if there exists a non-neg. fn $f_{XY}: \mathbb{R}^2 \to [0, \infty)$ s.t

$$F_{XY}(x, y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{XY}(s,t)\,dt\,ds$$

(f_{XY}: joint pdf)

① $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(s,t)\,ds\,dt = 1$

② $F_X(x) = \lim_{y\to\infty} F_{XY}(x, y)$

$$= \int_{-\infty}^{x}\int_{-\infty}^{\infty} f_{XY}(s,t)\,dt\,ds$$

$$= \int_{-\infty}^{x} f_X(s)\,ds$$

(f_X: pdf)

$\to f_X(s) = \int_{-\infty}^{\infty} f_{XY}(s,t)\,dt$ — Marginal pdf.

$\to f_Y(t) = \int_{-\infty}^{\infty} f_{XY}(s,t)\,ds$

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→ Eg – $F_{XY}(x,y) = 1 - e^{-x} - e^{-y} + e^{-(x+y)}$, $x, y \geq 0$

$f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y)$

$= \frac{\partial}{\partial x}\left[e^{-y} - e^{-(x+y)}\right] = e^{-y}\frac{d}{dx}\left[1 - e^{-x}\right]$

$= e^{-y} \cdot e^{-x} = e^{-(x+y)}$, $x, y \geq 0$

$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,t)\,dt = \int_0^{\infty} e^{-(x+t)}\,dt$

$= e^{-x}\int_0^{\infty} e^{-t}\,dt$

$= e^{-x}$, $x \geq 0$

→ $Exp(1)$

$f_Y(y) = \int_0^{\infty} e^{-(s+y)}\,ds = e^{-y}$, $y \geq 0$

$P(X \in (1,5),\ Y \in (3,6)) = \int_3^6 \int_1^5 e^{-(x+y)}\,dx\,dy$

Jointly Gaussian RVs

(X, Y) → standard bivariate Gaussian Random vector

$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{[x^2 - 2\rho xy + y^2]}{2(1-\rho^2)}\right)$

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$(x, y) \in \mathbb{R}^2 \quad . \quad \rho \in [-1, 1]$ → correlation coeff.

$f_X(x)$, $f_Y(y)$ ~ Marginal $N(0,1)$ $\quad f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$

$X \sim f_X \qquad F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$

$F_X(x) = P(X \le x)$

(S_1, S_2) → log return $S_1 \to \log\left(\frac{P_t^1}{P_{t-1}^1}\right) X$

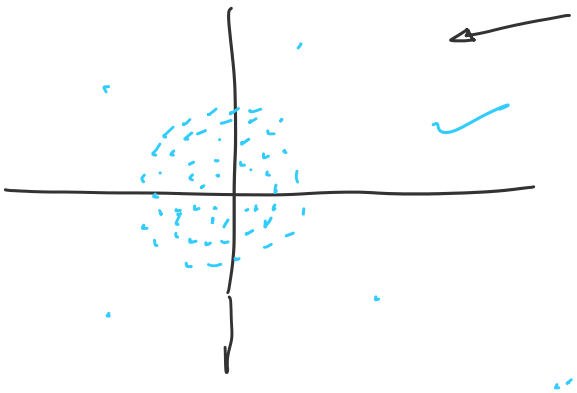
(X, Y) → Jointly Gaussian Random vector

$\quad\quad$ └ $S_2 \to \log\left(\frac{P_t^2}{P_{t-1}^2}\right) Y$

$N(0,1) \to X$ — [Channel] → $Y = X + Z$

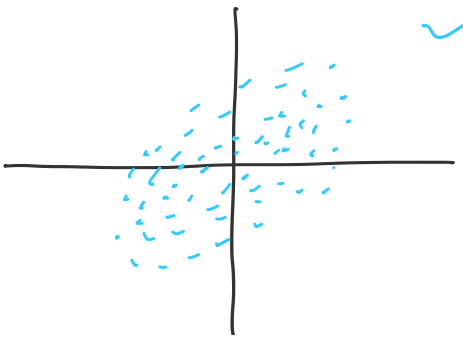
[Channel] ← [$Z \sim N(0, \sigma^2)$]

(X, Y) → Jointly Gaussian



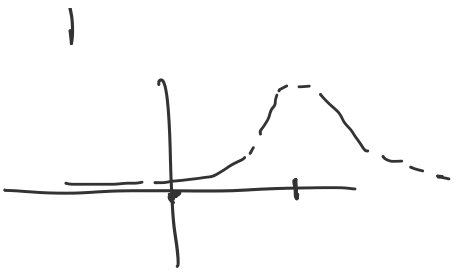
X, Y ind.

$X, Y \sim N(0,1)$



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(Height, weight)



$(\log(H), \log(W)) \rightarrow$ Jointly Gaussian RV

$\log(H) \sim \mu_H, \sigma_H$; $\log(W) \sim \mu_W, \sigma_W$



$\rho > 0$

Bivariate G.R.V

$$f_{HW}(x,y) = \frac{1}{2\pi\sigma_H\sigma_W\sqrt{1-\rho^2}} \exp\left(-\left[\left(\frac{x-\mu_H}{\sigma_H}\right)^2 + \left(\frac{y-\mu_W}{\sigma_W}\right)^2 + \dots\right]\right)$$

$$f_H \sim N(\mu_H, \sigma_H^2), \quad f_W \sim N(\mu_W, \sigma_W^2)$$

Conditional pdf

$$f_{X|Y}(x|y) \triangleq \frac{f_{XY}(x,y)}{f_Y(y)}, \text{ where } f_Y(y) > 0$$

$f_{XY}(x,y)$ ← joint density (X,Y)

$f_Y(y)$ → marginal density of Y

$Y = y$

Discrete RV

$$P_{X|Y}(X=x|Y=y) = \frac{P_{XY}(x,y)}{P_Y(y)} = \frac{P(X=x, Y=y)}{P(Y=y)}$$

Independence → $F_{XY}(x,y) = F_X(x) \cdot F_Y(y)$

Jointly cont → $f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$

$\forall (x,y) \in \mathbb{R}^2$

$f_{X|Y}(x|y) = f_X(x)$, $f_Y(y) > 0$

CDF

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Eg → $F_{XY}(x,y) = 1 - e^{-x} - e^{-y} + e^{-(x+y)}$ $x,y \geq 0$

→ $f_{XY}(x,y) = e^{-(x+y)}$ $x,y \geq 0$

→ $f_X(x) = e^{-x}$, $f_Y(y) = e^{-y}$

→ $f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$ $\forall\, x,y$

→ X, Y ind. R.V.

Eg Jointly std. Gaussian R.V. → $\rho = 0$

$$f_{XY}(x,y) = \frac{1}{2\pi} \exp\left(-\frac{(x^2+y^2)}{2}\right) \quad x,y \in \mathbb{R}^2$$

$$f_X(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad f_Y(y) = \frac{e^{-y^2/2}}{\sqrt{2\pi}}$$

$\rho = 0 \rightarrow f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$, X, Y ind.

$F_{XY}(x,y) = 1 - e^{-x} - e^{-y} + e^{-(x+y)}$ $x,y \geq 0$

X, Y ind ?

F_X, F_Y $(1-e^{-x})$, $(1-e^{-y})$

$F_{XY}(x,y) \overset{?}{=} F_X(x) \cdot F_Y(y)$

$P(X \leq x)$

$F_X(x) = \lim_{y \to \infty} F_{XY}(x,y) = 1 - e^{-x}$, $x \geq 0$ → $P(X \leq x, Y \leq y)$

$F_Y(y) = \lim_{x \to \infty} F_{XY}(x,y) = 1 - e^{-y}$, $y \geq 0$

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Eg → E-commerce company

$X \to$ norm. eng. score $\in (0,1)$

$Y \to$ norm. purchase value $\in (0,1)$

$$f_{XY}(x,y) = 1 + (2x-1)(2y-1), \quad x \in (0,1), \; y \in (0,1)$$

$f_{XY} \geq 0$

$$\int_0^1 \int_0^1 f_{XY}(x,y)\, dx\, dy = 1$$

Marginal densities

$$f_X(x) = \int_0^1 f_{XY}(x,y)\, dy$$

$$= \int_0^1 [1 + (2x-1)(2y-1)]\, dy$$

$$= \int_0^1 1 \cdot dy + (2x-1) \int_0^1 (2y-1)\, dy$$

$$= 1 + (2x-1) \cdot \left[2 \cdot \int_0^1 y\, dy - \int_0^1 1 \cdot dy \right]$$

$$= 1, \quad x \in (0,1)$$

$$2 \cdot \left[\frac{y^2}{2}\right]_0^1 - 1$$

$$2 \cdot \frac{[1]}{2} - 1 = 0$$

$\to f_X(x) = 1, \quad x \in (0,1) \quad X \sim \text{Unif}(0,1)$

$\to f_Y(y) = 1, \quad y \in (0,1) \quad Y \sim \text{Unif}(0,1)$

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Are X, Y ind? $f_{XY}(x,y) \overset{?}{=} f_X(x) \cdot f_Y(y)$

$1 + (2x-1)(2y-1) \neq 1 \cdot 1$

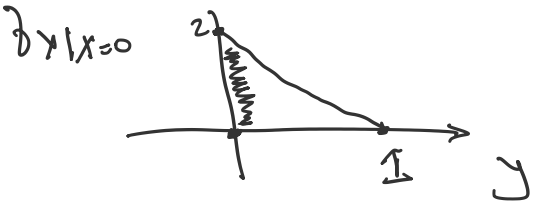
Cond. density of Y given $X \approx x$

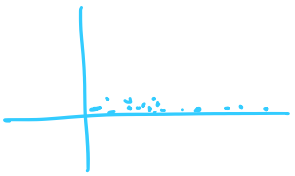
$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)}$$

$x=0$: $1 + (-1)(2y-1) = 2 - 2y$

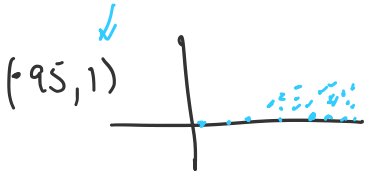
$$= 1 + (2x-1)(2y-1), \quad 0 < y < 1$$

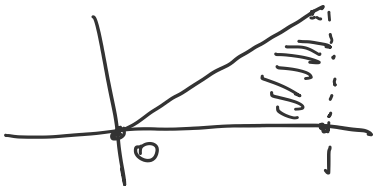
$x = 0 \rightarrow f_{Y|X}(y|0) = 2(1-y)$



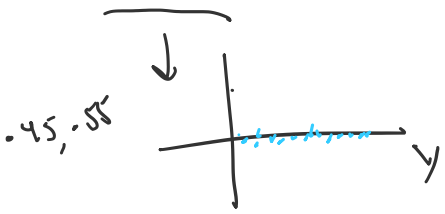


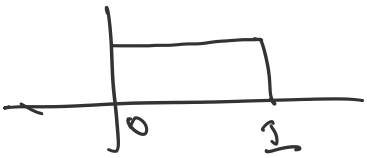
$x = 1 \rightarrow f_{Y|X}(y|1) = 2y, \quad 0 < y < 1$





$x = \frac{1}{2}$ $\quad f_{Y|X}(y|\tfrac{1}{2}) = 1 \quad Y \sim \text{unif}[0,1)$





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PROPERTIES of Expectation

$$E(X) = \int_{-\infty}^{\infty} x \, f_X(x)\, dx$$

$$\hookrightarrow \sum_{k \in S_X} k \cdot P_X(k)$$

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx$$

$$\hookrightarrow \sum_{k \in S_X} g(k) \cdot P_X(k)$$

① Linearity of Expectation

ⓐ $E(a \cdot X) = a \cdot E(X)$

ⓑ $E(X+Y) = E(X) + E(Y)$

↳ Does Not require X, Y to be ind.

② Independence X, Y ind. random variables

$$E(X \cdot Y) = E(X) \cdot E(Y)$$

Ind $\longrightarrow$ $E(XY) = E(X) \cdot E(Y)$ $\not\longleftarrow$

Eg Balls in Bins $\rightarrow$ m balls, n bins

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$\mathbb{E}[\#\text{ empty bins}]$ ✓

Z = # empty bins. $Z \in \{0, 1, 2 \ldots n\}$

$P_Z = P(Z = i)$ ✗

$Z = \sum X_k$ $\qquad \mathbb{E}(Z) = \sum \mathbb{E}(X_k)$

$X_k = \mathbb{1}_{\{\text{Bin } k \text{ is empty}\}} \rightarrow \begin{cases} 0 \rightarrow \text{Bin } k \text{ is not empty} \\ 1 \rightarrow \text{Bin } k \text{ is empty} \end{cases}$

$$Z = \sum_{k=1}^{n} X_k$$

$\mathbb{E}(X_k) = P(\text{Bin } k \text{ is empty})$

$= \left(\frac{n-1}{n}\right)^m$ $\qquad k \in \{1, 2 \ldots n\}$

$\mathbb{E}(Z) = n \cdot \left(\frac{n-1}{n}\right)^m$

$m \rightarrow$ Balls

$n \rightarrow$ Bins

$Prob(\text{Bin } k \text{ is empty})$

$= Prob\left(\bigcap_{i=1}^{m} \{\text{Ball } i \text{ skips Bin } k\}\right)$

$= \prod_{i=1}^{m} P(\text{Ball } i \text{ skips Bin } k)$

$\left(\frac{n-1}{n}\right) = \left(\frac{n-1}{n}\right)^m$

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Eg

$$F_X(x) = 1 - e^{-x}, \quad x \geq 0$$

$$f_X(x) = \frac{\partial}{\partial x} F_X(x) = e^{-x}, \quad x \geq 0$$

$\hookrightarrow X \sim Exp(1)$

$$F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

$$f_X(x) = \frac{\partial}{\partial x}\left[1 - e^{-\lambda x}\right] = \lambda \cdot e^{-\lambda x}, \quad x \geq 0$$

$\hookrightarrow Exp(\lambda)$

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